

Addis Ababa Science and Technology University

College of Engineering

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Internet Programming I (SWEG3107)

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Chapter One:

Introduction to Internet and Web Technologies

Contents

- Introduction
- World Wide Web (WWW)
- Uniform Resource Locators (URLs)
- Hyper Text Transfer Protocol (HTTP)
- Webpage Design Tools
  - Client Side Scripting; HTML, CSS, JavaScript
  - Server Side Scripting; PHP

Introduction (network)

- A **network** is a group of connected computers that are able to send data to each other.

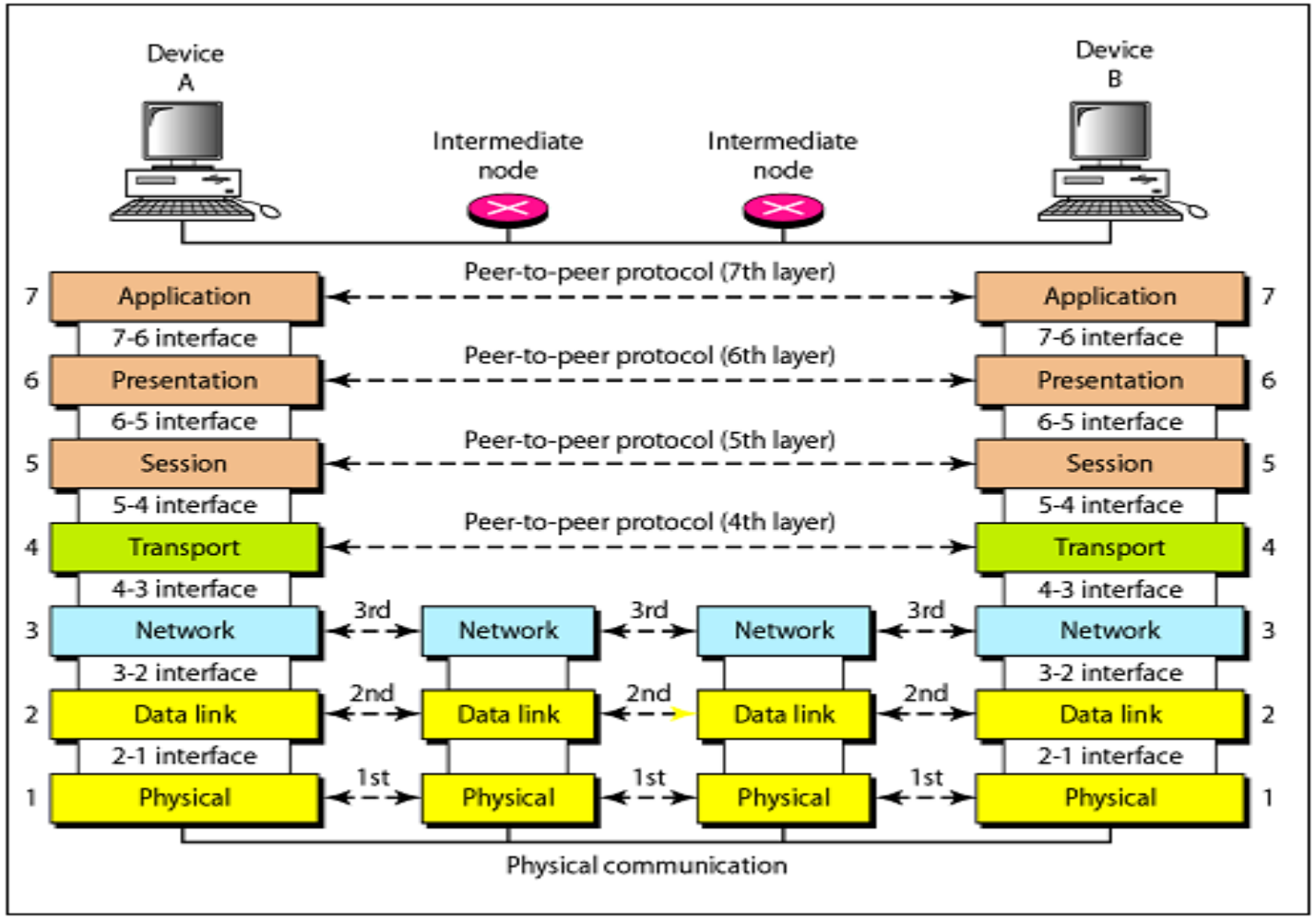


Fig: Communication & Interfaces in the OSI model

Introduction…

- The Application Layer, Transport Layer, and Network Layer are all integral components of network communication, each serving distinct but interconnected roles in the process of data transfer.

- **Application Layer:** is the topmost layer of the OSI model and is responsible for providing network services directly to end-user applications.
  - It facilitates user interactions with software applications, such as web browsers, email clients, and file transfer programs.

Introduction…

- **Transport Layer**: ensures reliable or unreliable delivery of data between applications running on hosts. It manages end-to-end communication, error detection, and flow control.
  - E.g., TCP (Transmission Control Protocol) for reliable communication and UDP (User Datagram Protocol) for faster, connectionless communication.

Introduction…

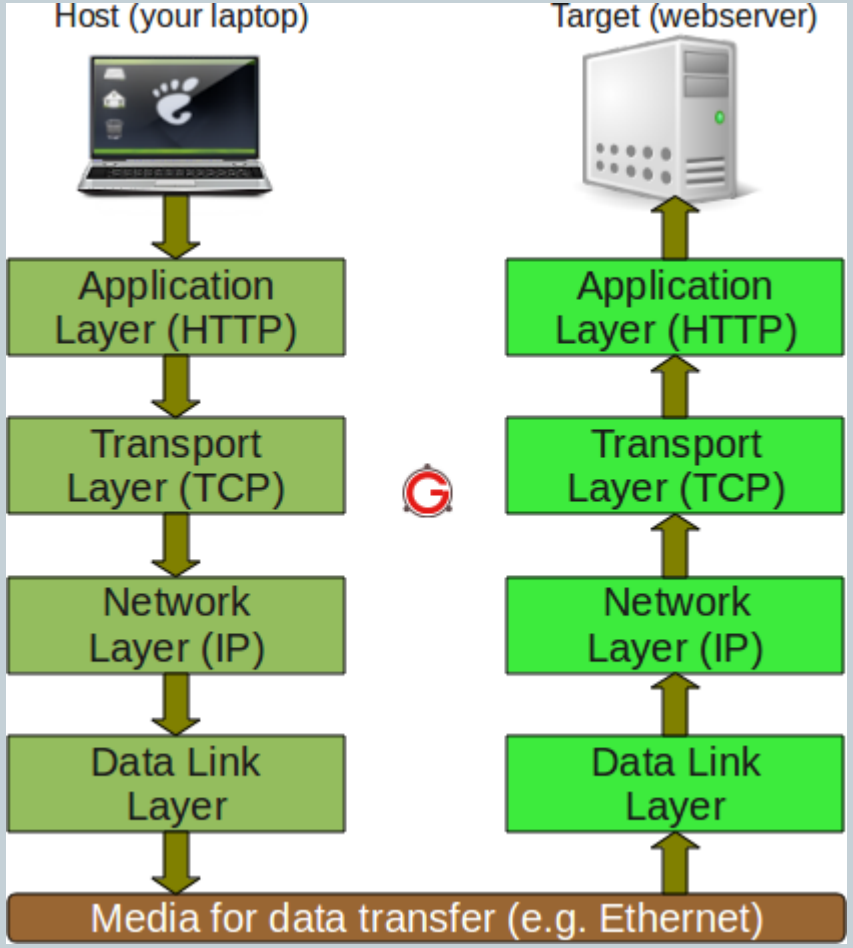
- **Network Layer:** is responsible for routing packets across multiple networks and managing logical addressing (IP addresses).
  - It determines the best path for data to travel from the source to the destination.
  - E.g., IP (Internet Protocol), ICMP (Internet Control Message Protocol), and various routing protocols like RIP and OSPF.

Introduction…

- **Internet:** it is a global network of interconnected computers and devices that communicate with each other using standardized protocols.
  - Network of networks
  - It uses standard internet protocol suite (TCP/IP) to connect billions of computer users worldwide.
  - It is set up by using cables such as optical fibers and other wireless and networking technologies.
  - At present, an internet is the fastest mean of sending or exchanging information and data between computers across the world.

Introduction…

- TCP/IP layer communication involves an encapsulation process on the sending side and a decapsulation process on the receiving side, where data passes through the Application, Transport, Internet, and Network Access layers.



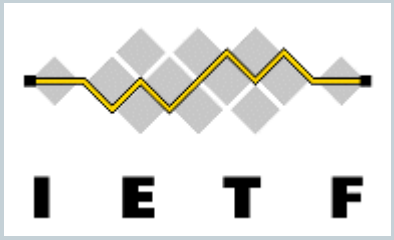
Introduction…

Several Applications of Internet

- **Communication(**Email for Instant communication, Messaging Apps to facilitate real-time messaging and collaboration, Video Conferencing to enable virtual meetings and webinars.
- **Information Access** (Search Engines: Google and Bing allow users to find information quickly, Online Encyclopedias, News Websites: Real-time news updates from platforms like BBC, CNN, and Reuters.

Introduction…

- **Social Networking**
- **E-Commerce** (Online Shopping, Digital Payment Systems)
- **Education** (E-Learning Platforms, Virtual Classrooms)
- **Internet Engineering Task Force (IETF):** internet protocol standards (Develops and promotes voluntary internet standards, particularly protocols that comprise the internet.)



Introduction…

- **Internet Corporation for Assigned Names and Numbers (ICANN):** decides top-level domain names or Oversees domain name registrations and IP address allocation.



- **World Wide Web Consortium (W3C):** Focuses on standards for web technologies to ensure the long-term growth of the web.

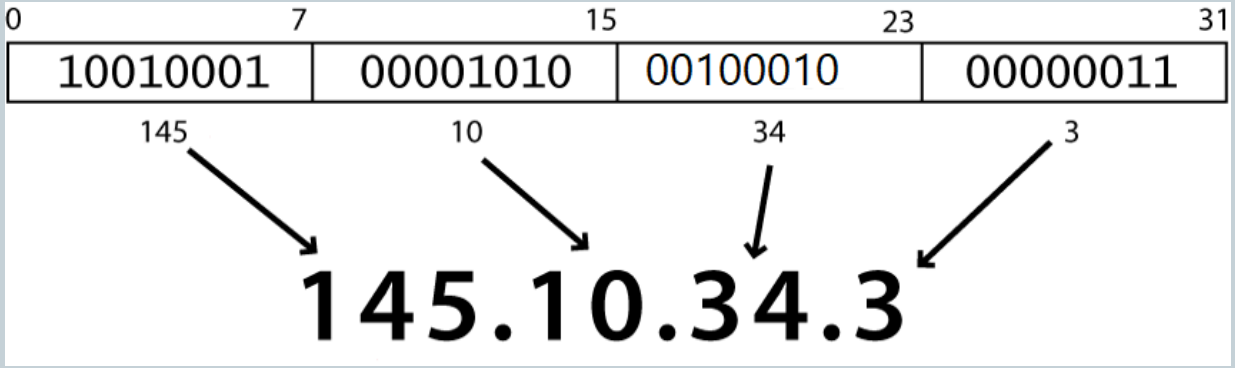


Home Reading Assignment

- Identify standards provided by IETF and W3C
- History of Internet
- Computer Networking Models (i.e., OSI and TCP/IP)
- Data Communication and Computer Network

Introduction… (IP Address)

- An IP address (Internet Protocol address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.
- Each machine is identified by a unique 4-byte number



- IP addresses are normally written as four numbers separated by a period, like this: 213.55.96.152

Introduction… (Domain Name Server)

- IP-addresses are hard to remember, Names are easier to remember than a 12 digit number.
- A **Domain Name** is simply a textual label that corresponds to a numerical IP address. E.g., www.google.com is a domain name
- When you address a web site, like http://www.google.com, the name is translated to a number by a Domain Name Server (DNS).
- When a new domain is registered together with a TCP/IP address, DNS servers all over the world are updated with this information

Introduction… (Domain Name Server)

DNS name structure

- DNS has a hierarchical structure that organizes domain names in a way that is both logical and manageable.
- **Hierarchical Structure:** DNS names are organized hierarchically from right to left, with each level separated by a dot (.)
- **Root Level:** The top of the DNS hierarchy is represented by an empty label (often depicted as a dot). This is the starting point for all domain names.

Introduction… (Domain Name Server)

- **Top-Level Domains (TLDs):** Directly below the root are TLDs, which represent the highest level in the DNS hierarchy.
  - E.g.,
    - Generic TLDs (gTLDs): .com, .org, .net, .info, etc.
    - Country Code TLDs (ccTLDs): .et (Ethiopia), .uk (United Kingdom), .jp (Japan), etc.
- **Second-Level Domains (SLDs):** These are directly to the left of the TLD and often represent the organization or entity owning the domain.
  - E.g., in "example.com," "example" is the second-level domain.

Introduction… (Domain Name Server)

- **Subdomains:** Subdomains are additional levels of hierarchy that can be created under a second-level domain. For instance, in "blog.example.com," "blog" is a subdomain of "example.com.“ E.g.,

Consider the domain name: www.blog.example.com

- **Root:** . (implicit, not shown)
- **TLD:** .com
- **Second-Level Domain:** example
- **Subdomain:** blog
- **Host Name:** www (a common subdomain representing a web server)

- Labels can be up to 63 characters long and are case **insensitive.**

Introduction… (World Wide Web)

- The World Wide Web (WWW or Web) is often confused with the Internet. The Web didn't exist until the 1980s
  - It is a collection of websites or web pages stored in web servers and connected to local computers through the internet.
  - It allows users to view and interact with a vast array of information using web browsers.
  - In 1989 Tim Berners-Lee at CERN created a set of technologies that allowed information on the Internet to be linked together through the use of links, or connections in documents
    - http://www.w3.org/History/1989/proposal.html

Introduction… (World Wide Web)

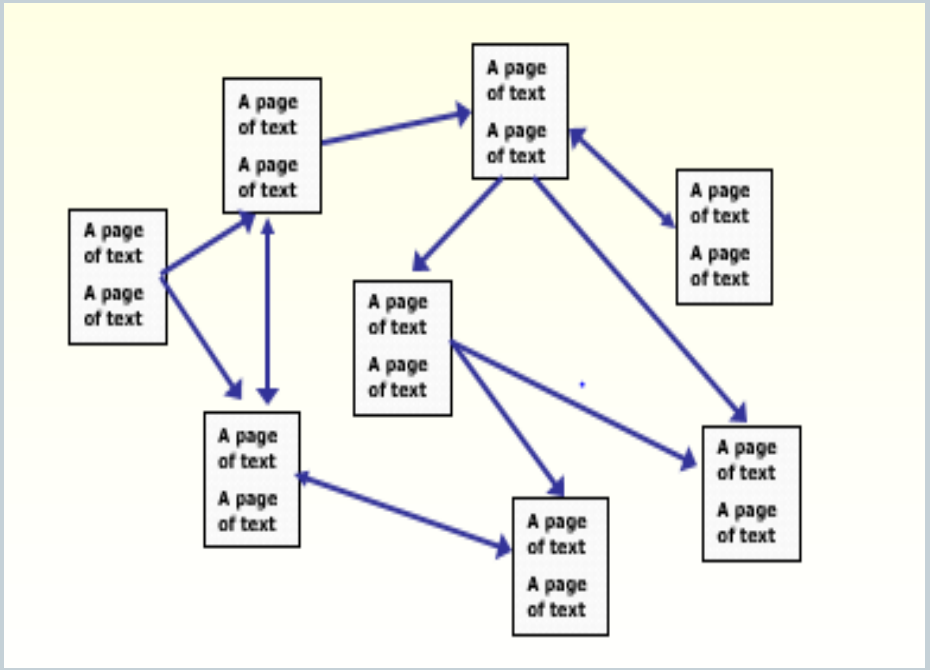
- WWW is **one of services** running over the Internet.
- The language used to **write** these documents with **links** is HTML-tells browsers how to display the information.
- The web is “governed” by the w3c (world wide web consortium) (www.w3.org)
- **Web Page:** Content on the web is typically formatted in and can include text, images, videos, and other multimedia.
- Web Pages are stored on web server and can be viewed using a web **browser**.
- *Collection of linked web pages on a web server is known as* ***website.***

Introduction… (Website)

- **A website** is a collection of related web pages that are typically identified by a common domain name and hosted on at least one web server.
- A collection of resources:
  - Web pages (static / dynamic)
  - Media files (images, animations, sound, …)
  - Style files (CSS)
  - Documents
  - Has a globally unique **name**
    - E.g. www.aastu.edu.et

Introduction… (Hypertext)

- It is a system of organizing and linking text and other media in a way that allows users to navigate through information easily.
- Non-linear reading and writing
  - embedded links to text in other documents
  - allows non-sequential browsing



Introduction… (Uniform Resource Locators (URL))

- A URL is a specific type of Uniform Resource Identifier (URI) that provides a means to locate a resource on the internet.
  - is used to address a document on the Web.
- The exact address of a resource on the web.
- A full Web address is like:
  - http://www.scss.tcd.ie/Owen.Conlan/php/xpath/index.html

Introduction… (Uniform Resource Locators (URL))

Components of a URL

1. Scheme: This indicates the protocol used to access the resource.

   Common schemes include:

   - http (Hypertext Transfer Protocol)
   - https (HTTP Secure)
   - ftp (File Transfer Protocol)

2. Host: This is the domain name or IP address of the server hosting the resource.

   E.g., in www.example.com, "example.com" is the host.

Introduction… (Uniform Resource Locators (URL))

Components of a URL..

3. Port (optional): This specifies the port number on the server. If omitted, default ports are used (e.g., 80 for HTTP and 443 for HTTPS).
4. Path: This indicates the specific location of the resource on the server.

   E.g., /folder/page.html points to a specific file within a folder.
5. Query String (optional): This part starts with a question mark (?) and includes parameters that provide additional information to the server.

   E.g., ?id=123&sort=asc includes parameters like id and sort.

Introduction… (Uniform Resource Locators (URL))

Components of a URL..

6. Fragment (optional): This starts with a hash (#) and refers to a specific section within a web page. For example, #section1 would navigate to "section1" within the page.

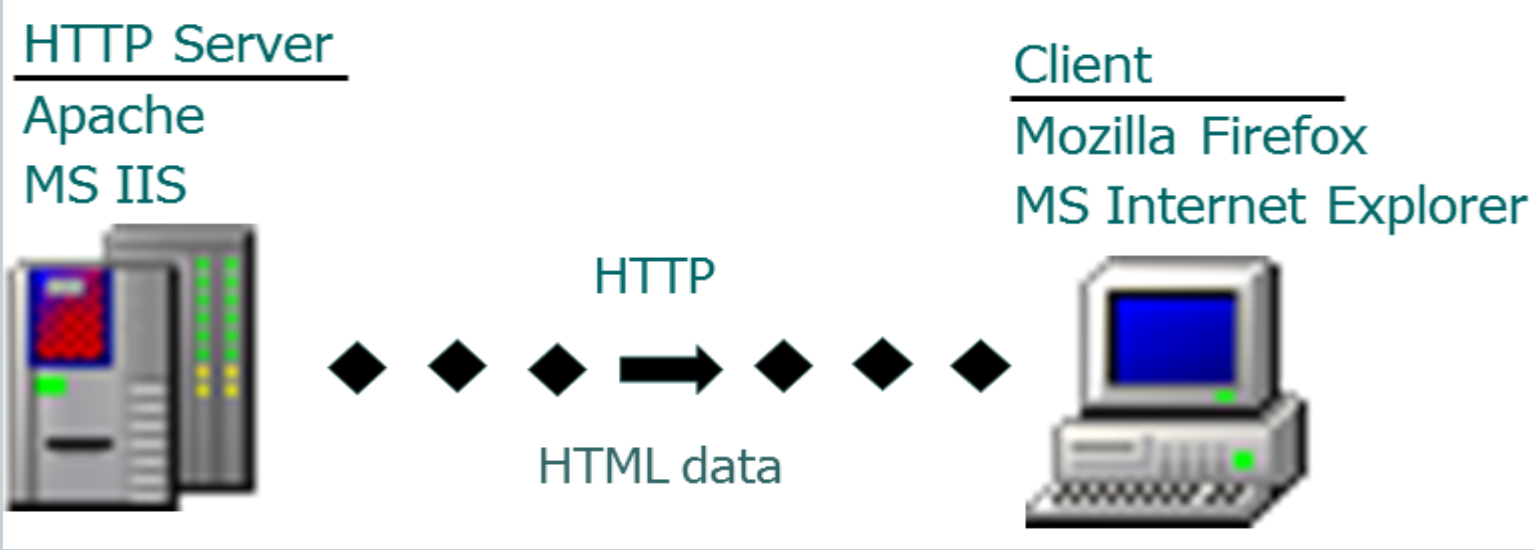
Example of a URL: **https://www.example.com:443/folder/page.html?id=123&sort=asc#section1**

- **Scheme:** `https`
- **Host:** `www.example.com`
- **Port:** `443` (default for HTTPS, often omitted)
- **Path:** `/folder/page.html`
- **Query String:** `?id=123&sort=asc`
- **Fragment:** `#section1`

Introduction… (Client-Server Architecture)

- **Client-server architecture** is a computing model that divides tasks between service providers (servers) and service requesters (clients).
- **Client -** initiate communication, request for info stored on server
- Clients can be desktop computers, mobile devices, or web browsers
- **A server** is a powerful computer or application that provides resources, data, or services to clients.
- Servers can host databases, applications, or websites.
- They process incoming requests and return the appropriate responses.

Introduction… (Client-Server Architecture)



Interaction Process

1. Request: The client sends a request to the server for specific data or services.

2. Processing: The server processes the request, which may involve retrieving data from a database or performing computations.

3. Response: The server sends the requested data or confirmation back to the client.

Introduction… (Multipurpose Internet Mail Extensions(MIME))

- A browser needs some way of **determining the format** of a document it receives from a Web server.
- MIME is used to **specify to browser the form of a file returned by server** (attached by the server to the beginning of the document).
  - different document formats require different rendering tools. E.g. text, image, video, …
- MIME defines various content types that specify the nature of the data being sent.
- Type specifications
  - Form: type/subtype
  - Examples: text/plain, text/html, image/gif, image/jpeg
  - Browser gets the type explicitly from the server

Introduction… (HyperText Transfer Protocol (HTTP))

- HTTP is an application layer protocol used for transmitting hypertext via the World Wide Web. It forms the foundation of data communication on the internet, enabling web browsers and servers to communicate with each other.
- Used to deliver all files and other data (collectively called *resources*) on the World Wide Web, whether they're HTML files, image files, query results, or anything else.
- Consists of two phases: Request and response. Each has two parts:
  - **header** contains information about the communication;
  - **body** contains the data of the communication if there is any

Introduction… (HTTP Request)

- **1. Request Phase Form:**
  a) HTTP method /path-to- resource HTTP/version-no
  b) Request Header fields
  c) blank line
  d) Message body

An example of the first line of a request:

- GET /cs.uccp.edu/degrees.html HTTP/1.1

Introduction… (HTTP Request…)

a) Request method**:** HTTP defines several methods (also known as verbs) to indicate the desired action:

- GET: Retrieve data from the server.
- POST: Send data to the server, often resulting in a change in state or side effects on the server.
- PUT: Update an existing resource or create a new resource.
- DELETE: Remove a specified resource.
- HEAD: Similar to GET, but retrieves only the headers, not the body.

b) HTTP headers: provide additional context about the request or response.

- They can include metadata such as content type, content length, caching policy

Introduction… (HTTP Request…)

Examples of headers include:

- **Content-Type**: Indicates the media type of the resource (e.g., text/html, application/json).
- **User-Agent**: Contains information about the client's software.

- Format: fieldname: value_of_field
- The header fields can be:
  - General headers: for general info, e.g. date
  - Request headers: info about browser, in request headers
  - Response headers: information about the server, for response headers
  - Entity headers: meta-information, used in both request & response headers

Introduction… (HTTP Request…)

- Common request fields:
  - Host: host name (specify host name, for HTTP 1.1)
  - Accept: text/plain (specify MIME type)
  - Accept: text/*
- Common response fields:
  - Content-length: 488
  - Content-type: text/html
- The order of the header field is insignificant

Introduction… (HTTP Request…)

c) blank line

- Separate the header from the body of the request.
- Requests that use the GET, HEAD, and DELETE methods **do not have bodies**. In these cases, the blank line signals are at the end of the request .

d) Message Body

- Used to carry an entity body
- May be divided into chunks and sent
- Optional, some cannot even have a message body
- Example of message body: A web page

Introduction… (HTTP Response)

2. Response Phase Form

a) Status line: includes HTTP version used, status code for the response, and short explanation of status code.

b) Response header fields: contain several lines of information about the response. The essential field of the header is **Content-type**.

c) Blank line

d) Response body

Introduction… (HTTP Response…)

HTTP Response – Example

HTTP/1.1 200 ok

Server: Microsoft-IIS/5.0

Content-Type: text/html

Accept-Ranges: bytes

Last-Modified: Tue, 13 Aug 2007 09:34:22 GMT

Content-Length: 240

```
<html>
<head>
<title>page titile</title>
</head>
<body>some content</body>
</html>
```

Introduction… (HTTP Response…)

- The status code is a three-digit integer, and the first digit identifies the general category of response:
  - **1xx** indicates an informational message only
  - **2xx** indicates success of some kind
  - **3xx** redirects the client to another URL
  - **4xx** indicates an error on the client's part
  - **5xx** indicates an error on the server's part

Introduction… (HTTP Response…)

- The most common status codes are:
  - **200 OK** The request succeeded, and the resulting resource (e.g. file or script output) is returned in the message body.
  - **301 Moved Permanently**
  - **302 Moved Temporarily**
  - **303 See Other** *(HTTP 1.1 only)* The resource has moved to another URL (given by the **Location:** response header), and should be automatically retrieved by the client.
  - **404 Not Found** The requested resource doesn't exist.
  - **500 Server Error** An unexpected server error. The most common cause is a server-side script that has bad syntax, fails, or otherwise can't run correctly

Introduction… (HTTP Response…)

- 1xx – request received, processing continues

  E.g.,

  - 100 Continue – tells client to continue with a request

- 2xx – success, action was successfully received, understood and accepted

  E.g.,

  - 200 Ok – request has succeeded
  - 202 Accepted – request accepted but not processed

- 3xx – further action must be taken to complete the request

  E.g.,

  - 302 Found – resource found but temporarily moved

Introduction… (HTTP Response…)

- 4xx – client error or invalid request

  E.g.

  - 400 Bad Request – couldn't understand request
  - 401 Unauthorized – request requires authorization
  - 403 Forbidden – client may not have access to the resource

- 5xx – server error occurred

  E.g.

  - 500 Internal Server Error – server encountered an unexpected error (error/bug with a server side script)
  - 505 HTTP Version Not Supported – server doesn't support the HTTP version

Introduction… (Website Designing Tools)

- Hypertext Markup Language (HTML): define the **content** of web pages
- Cascading Style Sheets (CSS): **stylistic** info for web pages
- JavaScript: to program the **behavior** of web pages
- Hypertext Processor (PHP): dynamically create pages on a web server
- Structured Query Language (SQL): interaction with databases

Home Reading Assignment

- DNS Name Structure
- Components of URL
- Request Phase Form
- Response Phase Form

Ended!

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Chapter One:

Introduction to Internet and Web Technologies